Software Requirements Specification (SRS)

Software Engineering Laboratory

Project Title: Online Food Ordering System

Version 1.0

Project Name	Online Food Ordering System
Version	1.0
Prepared By	Student Name :Adithya Kumaresan
Department	Information Technology
Date	15.07.26

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the functional and non-functional requirements of the Online Food Ordering System. The system aims to automate the process of browsing restaurants, placing food orders, making online payments, tracking deliveries, and collecting customer feedback. This document serves as a reference for developers, testers, and stakeholders throughout the software development life cycle.

1.2 Scope

The Online Food Ordering System is a web/mobile-based platform that connects customers with restaurants. It replaces manual phone-based ordering with an automated online system. The platform allows customers to browse restaurants and menus, place orders, pay online, track delivery status, and rate their experience. Restaurant owners can manage their menu items and incoming orders through a dedicated dashboard.

The system will support:

- Customer Registration and Login
- Restaurant and Menu Browsing
- Cart and Order Placement
- Online Payment Processing
- Order Tracking
- Ratings and Reviews
- Restaurant Menu and Order Management
- Admin Reports and Monitoring

1.3 Definitions

Term	Description
Customer	Registered user who browses restaurants and places food orders
Restaurant Owner	User who manages restaurant menu items and incoming orders
Admin	System administrator who oversees restaurants, users, and reports
Cart	Temporary collection of menu items selected by a customer before checkout
Order	Confirmed request placed by a customer for one or more menu items
Delivery Partner	Person responsible for delivering the order to the customer
Payment Gateway	Third-party service used to process online payments securely

1.4 References

- IEEE 830 Software Requirements Specification Standard
- Software Engineering by Ian Sommerville
- Software Engineering by Roger S. Pressman

2. Overall Description

2.1 Product Perspective

The Online Food Ordering System is a standalone web/mobile application that interacts with a centralized database and an external payment gateway.

Users include:

- Admin
- Restaurant Owner
- Customer
- Delivery Partner

2.2 Product Functions

- User Registration and Login
- Browse Restaurants and Menus
- Add/Edit/Delete Menu Items
- Add to Cart and Place Order
- Make Online Payment
- Track Order/Delivery Status
- Rate and Review Restaurants
- Manage Orders (Accept/Reject/Update Status)
- Generate Reports
- Change Password
- Logout

2.3 User Characteristics

Admin

- Manage users and restaurants
- Monitor orders and payments
- View reports

Restaurant Owner

- Add and update menu items
- Accept or reject orders
- Update order status

Customer

- Browse restaurants and menus
- Place orders and make payments
- Track orders and give ratings/reviews

2.4 Operating Environment

- Windows 10/11 and Linux
- Android/iOS (mobile app, optional)
- Java JDK 17 / Node.js
- MySQL Database
- Eclipse/IntelliJ IDEA/VS Code

- Apache Tomcat or Node/Express Server (for Web Application)

2.5 Design Constraints

- Java/Node.js Programming Language
- MySQL Database
- Secure token/password-based authentication
- Integration with a third-party payment gateway
- Maximum response time less than 3 seconds

3. Specific Requirements

3.1 Functional Requirements

FR1 User Registration and Login

The system shall allow customers and restaurant owners to register and authenticate using email/phone number and password.

Inputs / Actions:

- Name
- Email/Phone Number
- Password
- Role (Customer/Restaurant Owner)

Output:

- Registration Successful
- Login Successful
- Invalid Credentials

FR2 Restaurant and Menu Browsing

The system shall allow customers to browse a list of restaurants and view their menu items with prices and descriptions.

Inputs / Actions:

- Location/Search Keyword
- Category Filter

Output:

- List of Restaurants
- Menu Item Details

FR3 Menu Management

The restaurant owner shall be able to manage menu items available to customers.

Inputs / Actions:

- Add Item
- Update Item
- Delete Item
- Set Availability

Output:

- Menu Updated Successfully

FR4 Cart and Order Placement

The system shall allow customers to add items to a cart and place an order.

Inputs / Actions:

- Selected Menu Items
- Quantity
- Delivery Address

Output:

- Order Placed Successfully
- Order Summary

FR5 Online Payment

The system shall allow customers to pay for their order using online payment methods through a secure payment gateway.

Inputs / Actions:

- Payment Method
- Card/UPI/Wallet Details

Output:

- Payment Successful
- Payment Failed

FR6 Order Tracking

The system shall allow customers to track the real-time status of their order from preparation to delivery.

Inputs / Actions:

- Order ID

Output:

- Order Status: Placed / Preparing / Out for Delivery / Delivered

FR7 Ratings and Reviews

The system shall allow customers to rate and review a restaurant and the delivered order after completion.

Inputs / Actions:

- Rating (1–5 stars)
- Review Comment

Output:

- Review Submitted Successfully

FR8 Order Management (Restaurant Owner)

The restaurant owner shall be able to view, accept, reject, and update the status of incoming orders.

Inputs / Actions:

- Order ID
- Action (Accept/Reject)
- Status Update

Output:

- Order Status Updated

FR9 Reports

The system shall generate reports for administrative purposes.

Output:

- Sales Report
- Order History Report
- Top Rated Restaurants Report
- Revenue Report

3.2 Non-Functional Requirements

Performance

- Response time should be less than 3 seconds for all customer-facing operations.

Reliability

- System availability should be 99%.

Security

- Password authentication with encrypted storage.
- Role-based access control (Admin, Restaurant Owner, Customer).
- Secure handling of payment information via a certified payment gateway.

Usability

- User-friendly and responsive interface.
- Easy navigation for browsing and ordering.
- Clear error messages for invalid input.

Maintainability

- Modular design separating customer, restaurant, and admin modules.
- Well-documented source code.

Scalability

- Support up to 500 restaurants.
- Support 50,000 registered customers.
- Support concurrent order placement during peak hours.

4. External Interface Requirements

4.1 User Interface

- Login/Registration Screen
- Restaurant Listing and Menu Screen
- Cart and Checkout Screen
- Payment Screen
- Order Tracking Screen
- Ratings and Review Screen
- Restaurant Dashboard (Menu and Order Management)
- Admin Dashboard and Reports Screen

4.2 Hardware Interface

- Desktop/Laptop/Smartphone
- Internet Connectivity Device (Router/Mobile Data)

4.3 Software Interface

- Java/Node.js
- MySQL
- JDBC Driver / ORM
- Payment Gateway API (e.g., Razorpay/Stripe)
- Apache Tomcat / Node Server

5. System Features

Feature 1: Place Order

Inputs: Selected Menu Items, Quantity, Delivery Address

Outputs: Order Placed Successfully

Feature 2: Make Payment

Inputs: Order ID, Payment Method, Payment Details

Outputs: Payment Successful / Payment Failed

Feature 3: Track Delivery

Inputs: Order ID

Outputs: Current Order/Delivery Status

Feature 4: Rate and Review

Inputs: Order ID, Rating, Review Comment

Outputs: Review Submitted Successfully

6. Data Requirements

User Table

Field	Type
User_ID	Integer
Name	Varchar
Email	Varchar
Phone	Varchar
Password	Varchar (Encrypted)
Role	Varchar

Restaurant Table

Field	Type
Restaurant_ID	Integer
Name	Varchar
Address	Varchar
Owner_ID	Integer
Rating	Double

Menu_Item Table

Field	Type
Item_ID	Integer
Restaurant_ID	Integer
Name	Varchar
Price	Double
Category	Varchar
Availability	Boolean

Order Table

Field	Type
Order_ID	Integer
Customer_ID	Integer
Restaurant_ID	Integer
Order_Date	Date

Field	Type
Status	Varchar
Total_Amount	Double

Payment Table

Field	Type
Payment_ID	Integer
Order_ID	Integer
Amount	Double
Payment_Method	Varchar
Payment_Status	Varchar

Review Table

Field	Type
Review_ID	Integer
Order_ID	Integer
Customer_ID	Integer
Rating	Integer
Comment	Varchar

7. Assumptions

- Internet connection is available on the customer and restaurant end.
- Database server is operational at all times.
- Only registered users can place orders.
- Payment gateway services are functional and reliable.

8. Constraints

- Customers cannot place an order if the restaurant is closed or the item is unavailable.
- Duplicate user accounts with the same email are not allowed.
- Orders cannot be cancelled once out for delivery.
- Ratings can only be submitted after order delivery is confirmed.

9. Acceptance Criteria

The software will be considered successful if it:

- Authenticates valid users correctly.
- Allows restaurant and menu browsing without errors.
- Processes order placement and payment reliably.
- Provides accurate real-time order tracking.
- Allows customers to submit ratings and reviews.
- Allows restaurant owners to manage menus and orders.
- Generates reports without errors.
- Maintains data consistency and security.